

2 Pineapple plants contain the enzyme bromelain that breaks down proteins such as casein.

(a) (i) Name the bond between amino acids that is broken by this enzyme.

(1)

(ii) Name the process that breaks the bond between amino acids.

(1)

(b) The digestion of casein by bromelain was investigated.

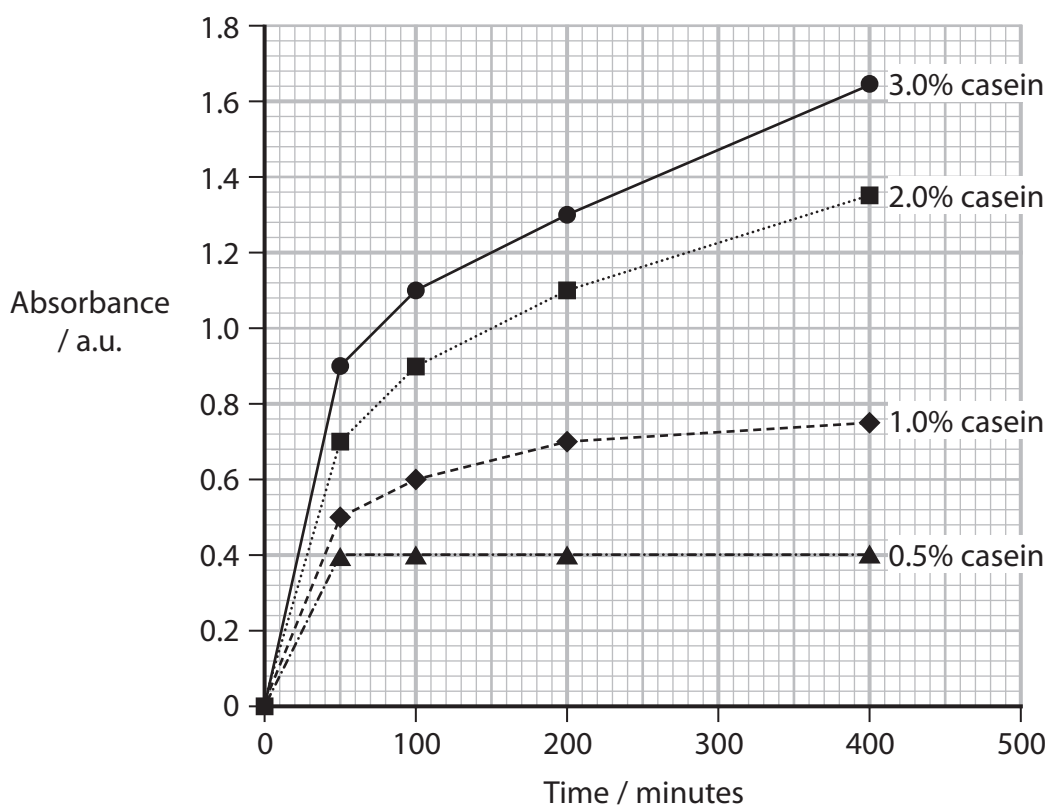
The amino acids in casein, a milk protein, were stained blue.

Four concentrations of this casein were used.

A colorimeter was used to measure the absorbance by the blue amino acids released in this reaction.

The more digestion that occurred, the more blue amino acids were released and the higher the absorbance reading on the colorimeter.

The graph shows the results of this investigation.



The initial rates of reaction can be obtained from this graph.

The table shows the initial rates of this reaction.

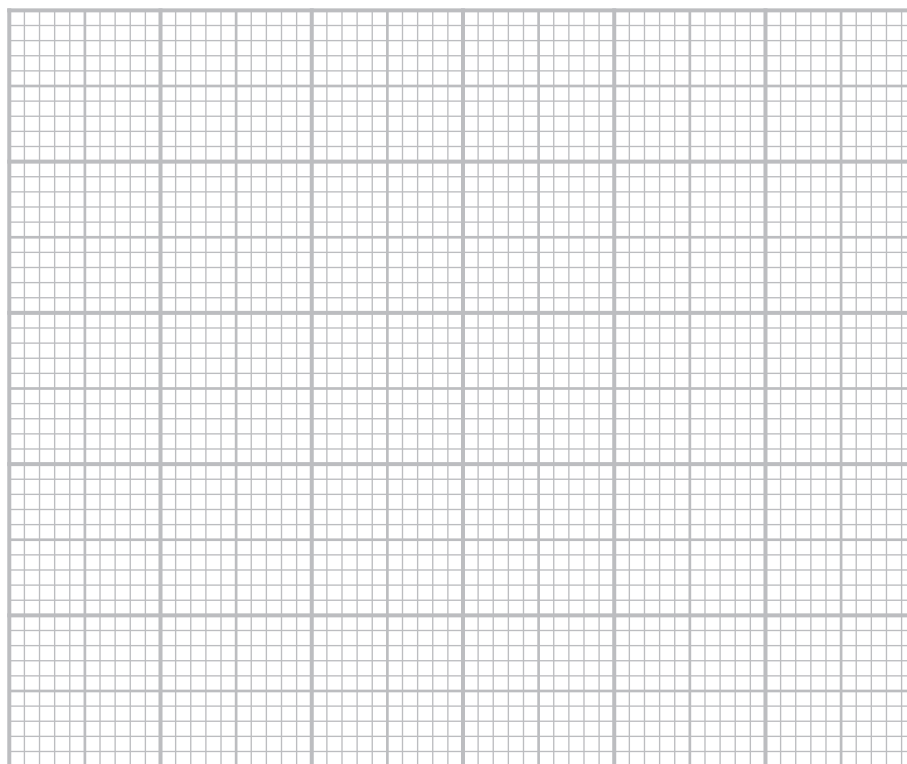
Substrate concentration (%)	Initial rate of reaction / a.u. min ⁻¹
0.5	0.008
1.0	0.010
2.0	0.014
3.0	

- (i) Complete the table by calculating the initial rate of reaction when a 3.0% concentration of casein was used.

(1)

- (ii) Plot a suitable graph to show the effect of substrate concentration on the initial rate of reaction.

(4)



- (iii) Calculate how many times faster the initial rate of reaction is for a casein concentration of 2.0% than a casein concentration of 0.5%.

Give your answer to 2 significant figures.

(1)

Answer

- (iv) Explain the shape of the graph you have plotted in (ii).

(3)

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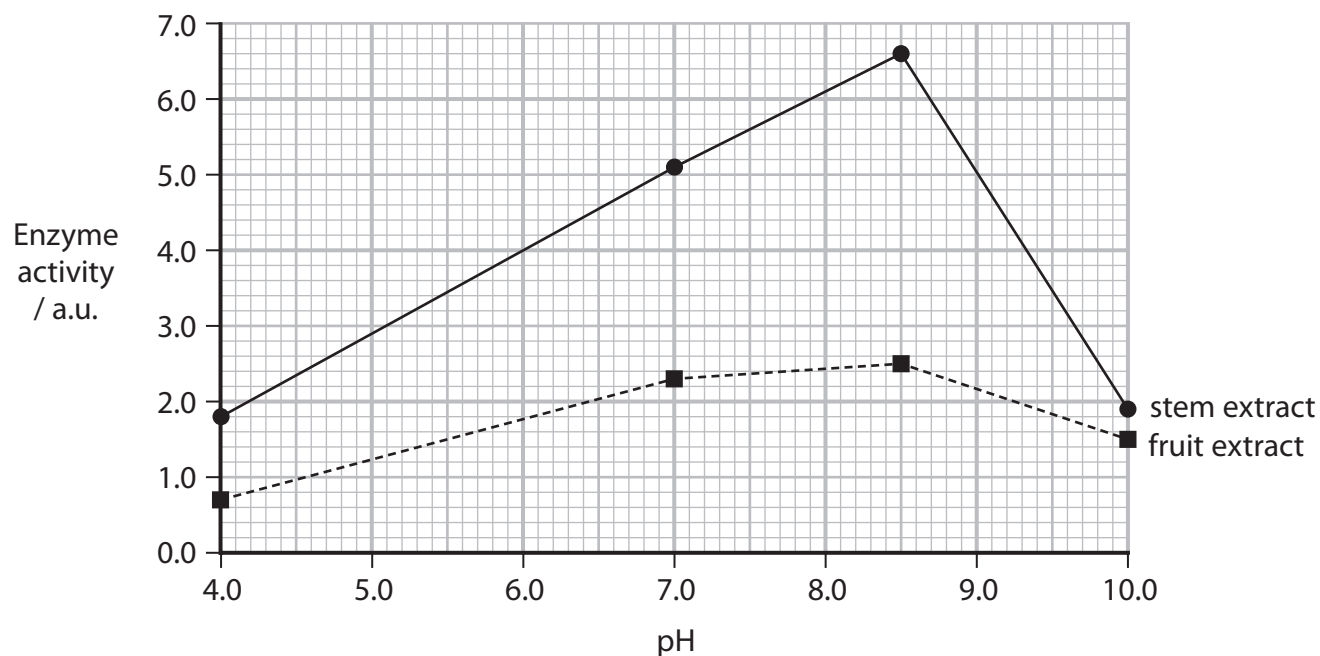


(c) Bromelain can be used to prevent apple juice from going brown.

Bromelain can be extracted from the stem and from the fruit of a pineapple.

The effect of pH on these extracts was investigated.

The graph shows the results of this investigation.



(i) Draw a suitable table to show the results of this investigation.

(3)



- (ii) Determine which of these extracts is most useful for preventing apple juice from going brown.

(3)

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(Total for Question 2 = 17 marks)

